📘 CHAPTER 4: DATA BACKUP & RECOVERY – DETAILED SUMMARY

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1. INTRODUCTION TO DATA BACKUP & RECOVERY

- **Backup**: The process of creating a duplicate copy of master and transaction data files.
- **Recovery**: The process of restoring lost or corrupted data to a consistent and usable state.
- **Purpose**: To protect data from loss, corruption, or disaster and ensure business continuity.
- **Backup copy (snapshot)** should ideally be **immutable**—cannot be altered after creation (protection against ransomware).

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2. TYPES OF DATA BACKUPS

Type	Description	Pros	Cons
Full Backup	Backs up **all data** from a server, database, or VM.	Complete protection.	Time-consuming, storage-heavy.
Incremental Backup	Backs up **only data changed since last backup** (full or incremental).	Fast, less storage.	Recovery requires full backup + all increments.
Differential Backup	Backs up data changed since **last full backup**.	Faster recovery than incremental.	Larger than incremental over time.

Incremental Backup Subtypes (from PDF):

- **Block-level backup**: Backs up only modified parts of files.
- **Byte-level backup**: Uses binary variation to identify changes.
- **Reverse incremental**: Most recent restore point is always a full backup.

- **Multilevel incremental**: Uses numbered backup levels (0 = full, n = changes since n–1).
- **Incremental forever**: After initial full, only increments sent; server reconstructs during restore.

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3. WHY DATA BACKUP IS NEEDED

Challenges (Risks) Leading to Data Loss:

1. **System failure** – Hardware/software crashes.
2. **Human error** – Accidental deletion/overwrite.
3. **Rogue insiders** – Malicious employees.
4. **Theft** – Physical device theft.
5. **Cyberattacks** – Especially ransomware.
6. **Natural disasters** – Floods, fires, earthquakes.

Reasons to Implement Backup:

- **Defend against cyberattacks** – Restore clean copies after ransomware.
- **Minimize downtime** – Quick recovery = less operational disruption.
- **Save money** – Avoid costs of data recreation and lost revenue.
- **Improve compliance** – Meet data protection regulations (e.g., GDPR).
- **Maintain customer trust** – Show commitment to data security.

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4. STORAGE MEDIA FOR BACKUP

Media	Description	Use Case
HDD	Magnetic storage, platters, read/write heads. Slower, mechanical.	High-capacity, cost-effective backups.
SSD	Flash memory, no moving parts, faster, more durable.	Fast backup/restore, performance-critical systems.

Media	Description	Use Case
RAID	Combines multiple disks for redundancy/performance.	Fault-tolerant storage systems.
USB Flash Drives	Portable flash memory with USB interface.	Small, portable backups.
Optical Drives	CDs, DVDs, Blu-rays.	Long-term archival, offline backups.

RAID Levels Explained:

- **RAID 0** – Striping (speed, no redundancy).
- **RAID 1** – Mirroring (duplicate data, fault tolerance).
- **RAID 5** – Striping with distributed parity (balance of speed & safety).
- **RAID 10** – Combines mirroring + striping (high performance & redundancy).

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5. DATA BACKUP METHODS

On-site vs. Off-site Storage:

- **On-site**: Backup stored locally. Fast access, but vulnerable to local disasters.
- **Off-site**: Backup stored elsewhere (e.g., cloud). Protects against site-wide disasters.

Online vs. Offline Backups:

- **Online**: Backup media is connected and available instantly. Faster but vulnerable to malware.
- **Offline**: Media is disconnected after backup. More secure against ransomware.

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6. DATA RECOVERY TECHNIQUES

Technique	Description
Software-Based Recovery	Uses tools to scan for and recover lost/deleted files.
Hardware-Based Recovery	Repairs physically damaged drives; done in cleanroom environments.
File Carving	Extracts data fragments without relying on file system metadata.
Remote Data Recovery	Specialists connect remotely to recover data over the internet.
Hardware Repair & Replacement	Fixing or replacing damaged components (e.g., read/write heads).
Forensic Data Recovery	Recovers data for legal purposes, preserving metadata and integrity.

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7. DATABASE BACKUP PLAN – STEP-BY-STEP

1. **Identify critical data** – Determine what must be backed up.
2. **Define recovery objectives**:
    - **RTO (Recovery Time Objective)**: Max acceptable downtime.
    - **RPO (Recovery Point Objective)**: Max acceptable data loss.
3. **Choose between online or offline backups** – Based on security and accessibility needs.
4. **Assign backup strategy** – Based on system capabilities (full, incremental, differential).
5. **Use data replication and backup tools** – Automate and ensure real-time copying where needed.

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8. KEY CONCEPTS TO MEMORIZE FOR TEST

- **Full vs. Incremental vs. Differential Backup**
- **RTO vs. RPO**
- **RAID 0, 1, 5, 10**

- **On-site vs. Off-site | Online vs. Offline**
- **Recovery techniques** (especially software-based, hardware-based, forensic)
- **Risks of data loss** (ransomware, human error, natural disasters)

🎯 **Test Tips:**

- **Scenario questions**: They may ask which backup type to use in a given situation.
- **RAID questions**: Know which RAID level offers what (speed, redundancy, or both).
- **RTO/RPO questions**: Understand which is about **time** and which is about **data loss**.
- **Recovery methods**: Know when to use hardware vs. software recovery.